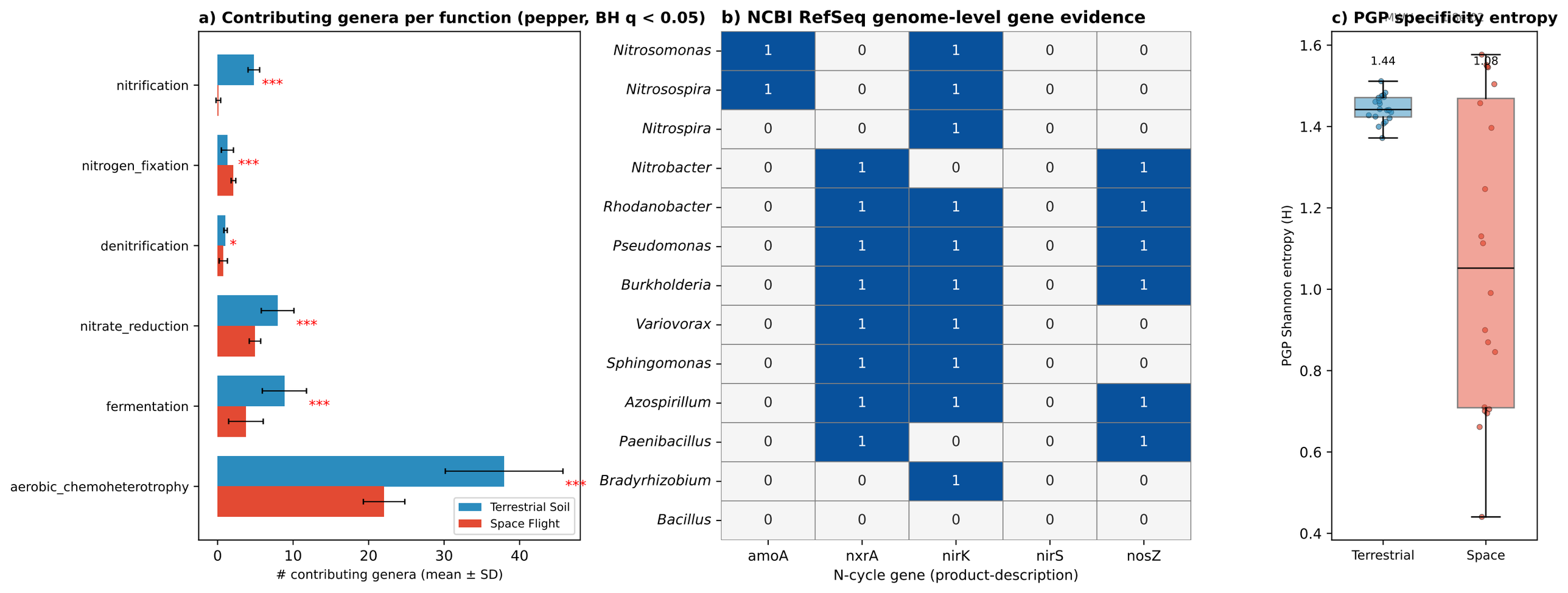
a) Contributing genera per function (pepper, BH q < 0.05)
nitrification
nitrogen_fixation
denitrification
nitrate_reduction
fermentation
aerobic_chemoheterotrophy
***
***
*
***
***
***
Terrestrial Soil
Space Flight
0
10
20
30
40
contributing genera (mean ± SD)
b) NCBI RefSeq genome-level gene evidence
Nitrosomonas 1 0 1 0 0
Nitrosospira 1 0 1 0 0
Nitrospira 0 0 1 0 0
Nitrobacter 0 1 0 0 1
Rhodanobacter 0 1 1 0 1
Pseudomonas 0 1 1 0 1
Burkholderia 0 1 1 0 1
Variovorax 0 1 1 0 0
Sphingomonas 0 1 1 0 0
Azospirillum 0 1 1 0 1
Paenibacillus 0 1 0 0 1
Bradyrhizobium 0 0 1 0 0
Bacillus 0 0 0 0 0
amoA
nxrA
nirK
nirS
nosZ
N-cycle gene (product-description)
c) PGP specificity entropy
PGP Shannon entropy (H)
1.6
1.4
1.2
1.0
0.8
0.6
0.4
1.44
1.08
Terrestrial
Space